UCL / ELECTRONIC & ELECTRICAL ENGINEERING

When can a robot **trust a map?**

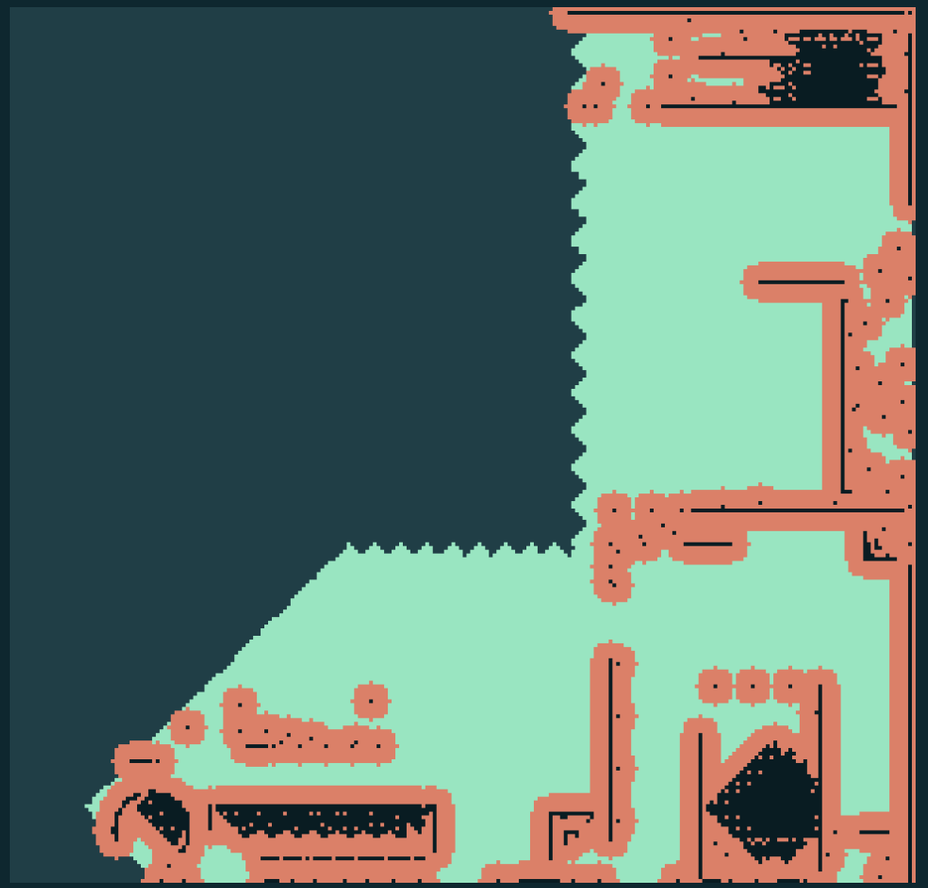
From noisy point clouds to traversability decisions.

I explored how a map can distinguish missing evidence, confirmed hazards and the cost of crossing observed terrain.

CHI ZOU

Year 4 project / Robotics mapping & system interfaces

5 cm grid resolution



Saved simulation state / step 50 of 50

13 reported scenes

50 simulation steps

S1: final report, Appendix D | S3: saved simulation arrays. Cover visual re-rendered from step 50.

A clear place in a larger robot system.

Project context: Swarm-Based Monocular Vision Scanning Robot, UCL Year 4.

01
Capture
Robot platform

>

02
Reconstruct
RGB to point cloud

>

03
Merge
Registration / fusion

>

04
2D map
My technical exploration

>

05
Plan
Frontier exploration

PERSONAL SCOPE

Feasibility-oriented point-cloud-to-2D-map exploration, with the theory, experimental implementation, quantitative comparisons and limitations documented in Appendix D.

THREE QUESTIONS I WORKED THROUGH

01 What evidence is enough to label a cell?

02 Which features should block motion or add cost?

03 How should the result be represented and handed over?

OWNERSHIP & VERSION BOUNDARY

The four-state simulation prototype and the later three-state cost-map study are distinct stages. Final mapping refactoring and system integration were completed by Zhuyu Dong; robot hardware, reconstruction, fusion and planning were team contributions.

Separate evidence from traversal difficulty.

The experimental map answers two questions: can a cell be considered, and how costly is it to cross?

01 / OBSERVE

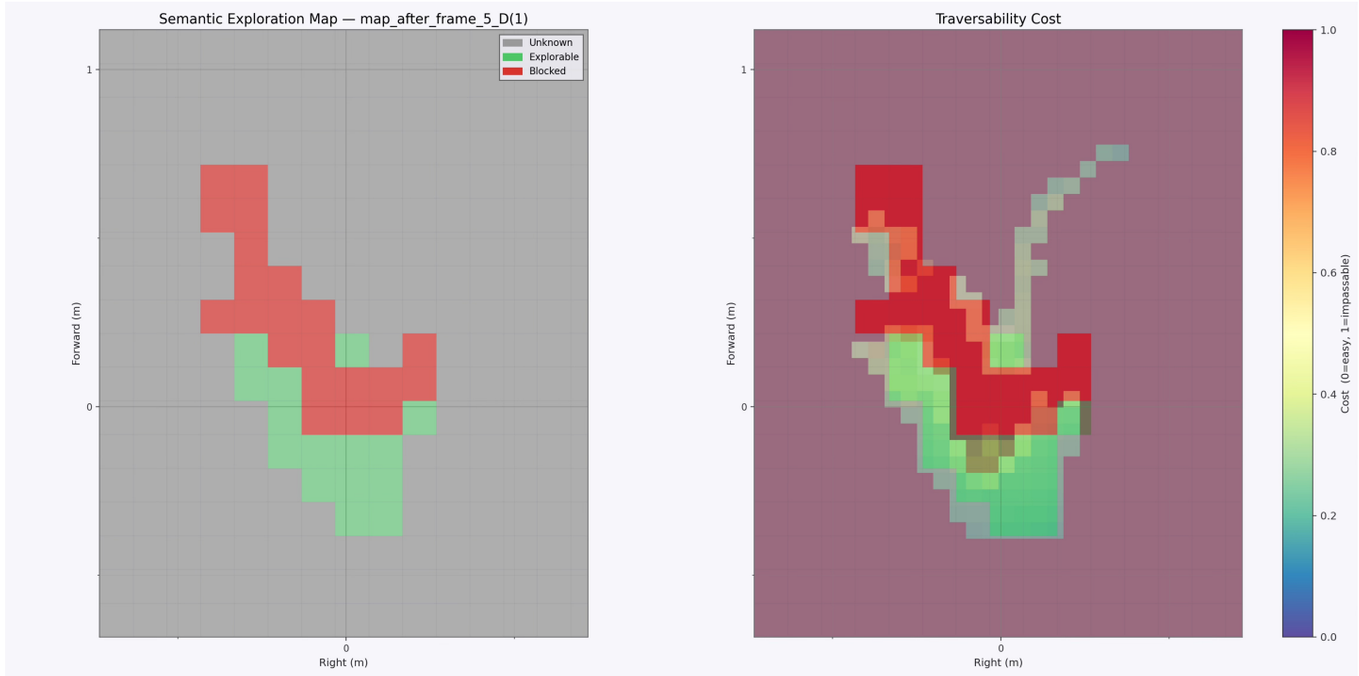
Extract a near-ground layer and check local observation support. Insufficient evidence remains UNKNOWN.

02 / BLOCK

Reserve hard blocking for obstacle-core evidence, supported step edges and configured inflation.

03 / COST

Within EXPLORABLE cells, combine slope, roughness, height variation and obstacle proximity.



frame_5_D / saved semantic map and continuous cost output

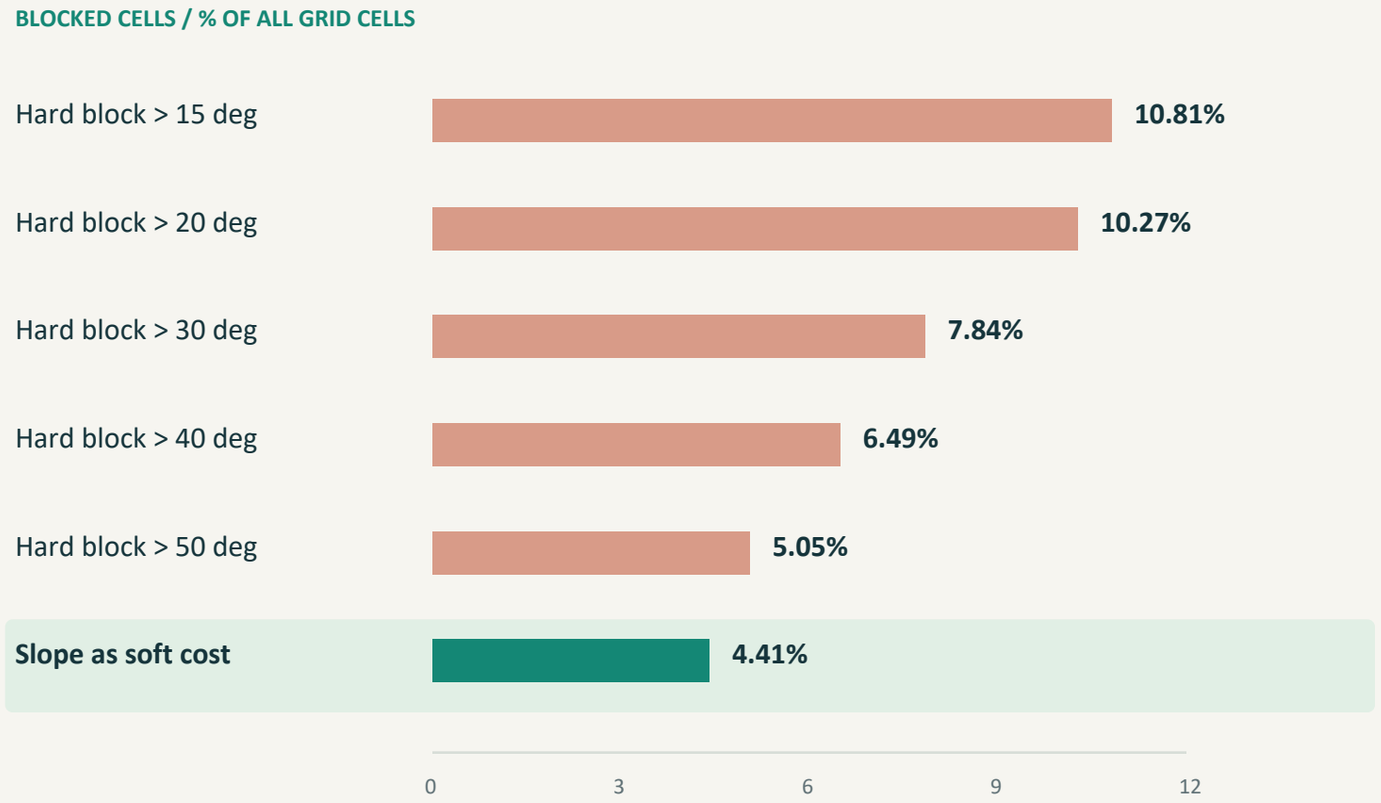
$$C = w_s S + w_r R + w_e E + w_p \exp(-d / \lambda)$$

Slope / roughness / elevation variation / obstacle distance

S1: Appendix D.1-D.2, PDF pp. 52-57 | S6: original frame_5_D figure. Equation describes the reported experiment.

A steep signal is not always a hard obstacle.

Ablation on frame_5_D: changing the role of slope changes the map available to a planner.



30 DEG HARD BLOCK VS SOFT COST

3.43 pp

lower blocked-cell share

EXPLORABLE cells increased from 8.11% to 11.44% in the same reported scene.

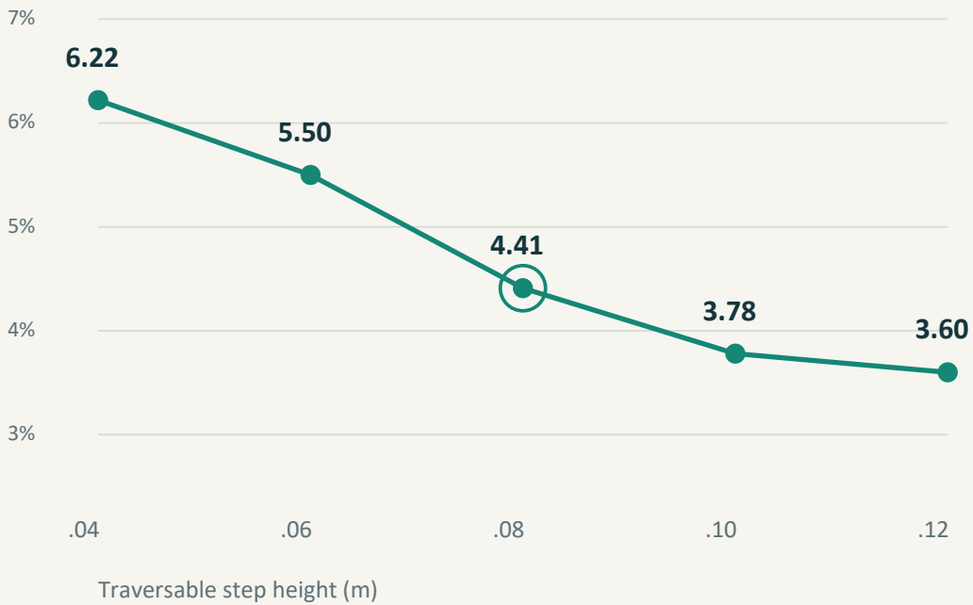
Interpretation: height noise can amplify local gradients. I retained slope as a graded penalty while requiring stronger evidence for a hard block.

S1: Table 7, PDF p. 59 | Single-scene class distribution, not an accuracy or false-positive-rate measurement.

Show the response, not just the chosen threshold.

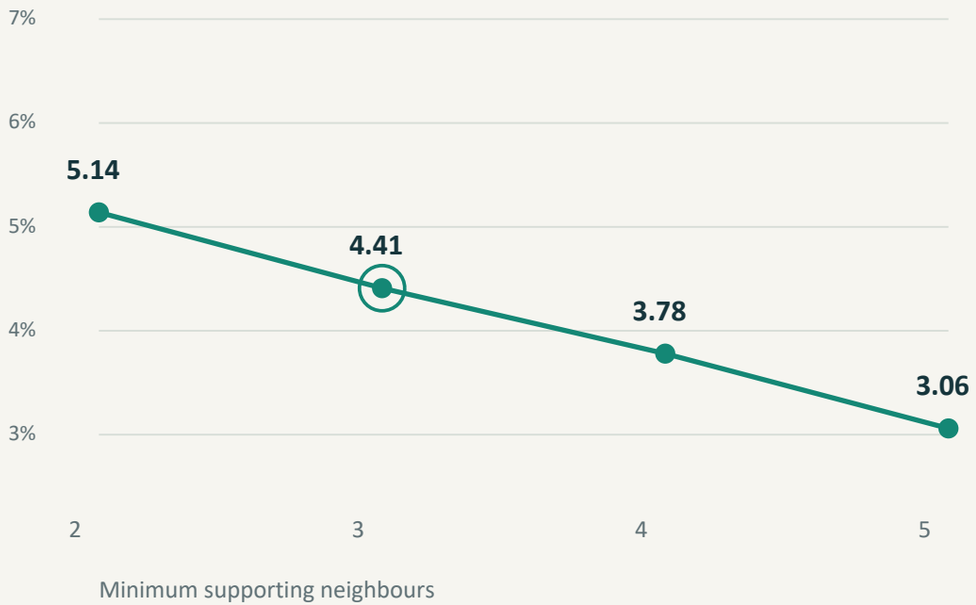
Two reported parameter sweeps on frame_5_D expose how strictness changes the blocked region.

STEP HEIGHT / BLOCKED SHARE



0.04 m to 0.12 m: BLOCKED falls from 6.22% to 3.60%. The reported default is 0.08 m.

NEIGHBOUR SUPPORT / BLOCKED SHARE



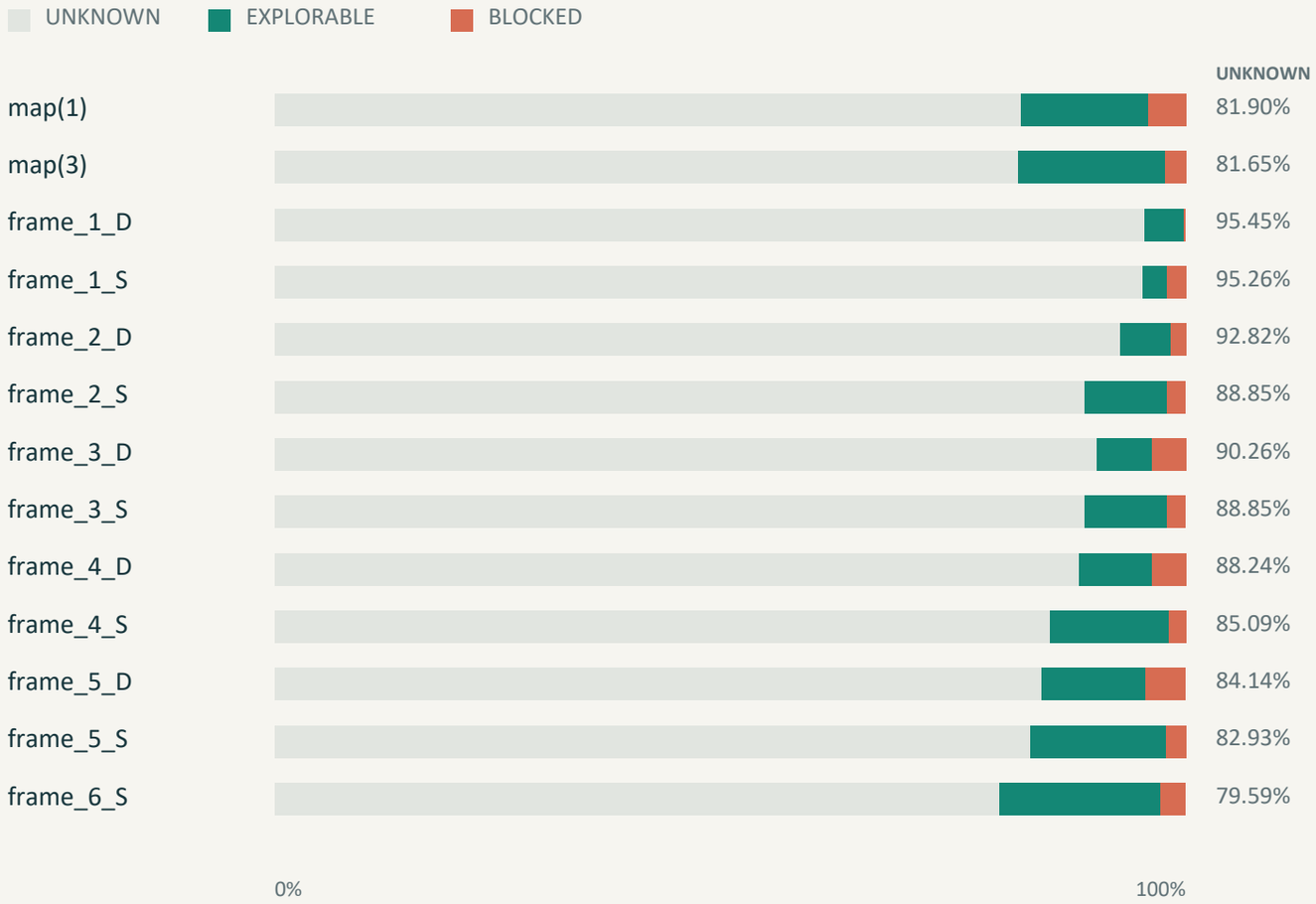
2 to 5 neighbours: BLOCKED falls from 5.14% to 3.06%. The reported default is 3.

Circled points mark defaults. UNKNOWN changes little; these controls mainly affect step-edge classification.

S1: Tables 5-6, PDF pp. 58-59 | Recorded settings only; connecting lines are visual guides, not extra measurements.

Unknown is a state worth preserving.

Thirteen reported scenes reveal both the structure of the output and the limit of the available observations.



88.24%

median UNKNOWN share
across the 13 reported scenes

SPARSE CASE / FRAME_1_D

95.45% UNKNOWN. Missing evidence is made explicit instead of silently being treated as free space.

RICHER CASE / FRAME_6_S

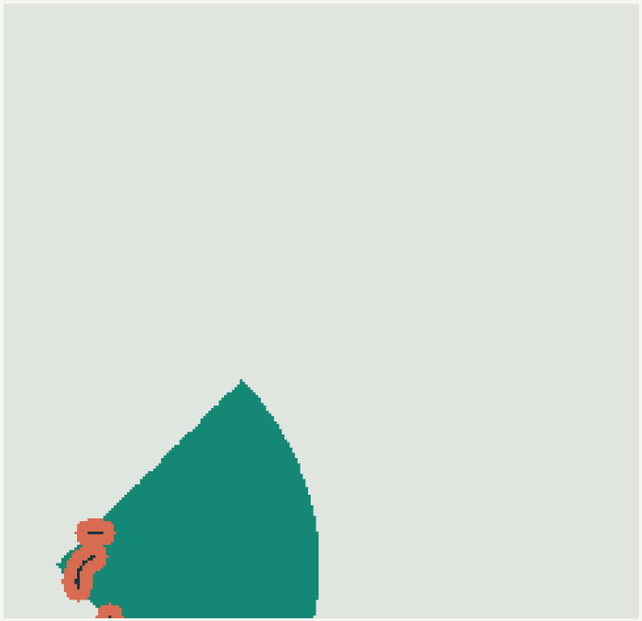
17.68% EXPLORABLE. More observed terrain gives the classifier more usable support.

S1: Table 4 and limitations, PDF pp. 58-59 | No external ground truth; this evaluates output consistency, not accuracy.

Replay the evidence, not just the final map.

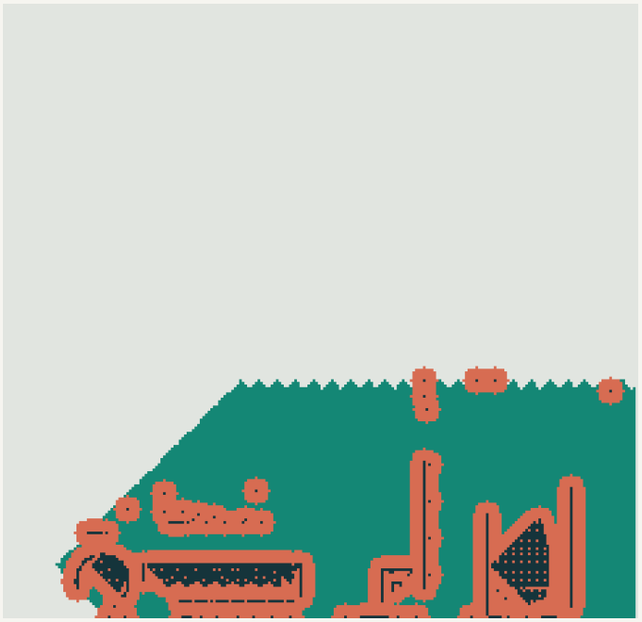
Saved four-state prototype output. Fixed grid and identical view bounds across all 50 steps.

UNKNOWN FREE OCCUPIED INFLATED



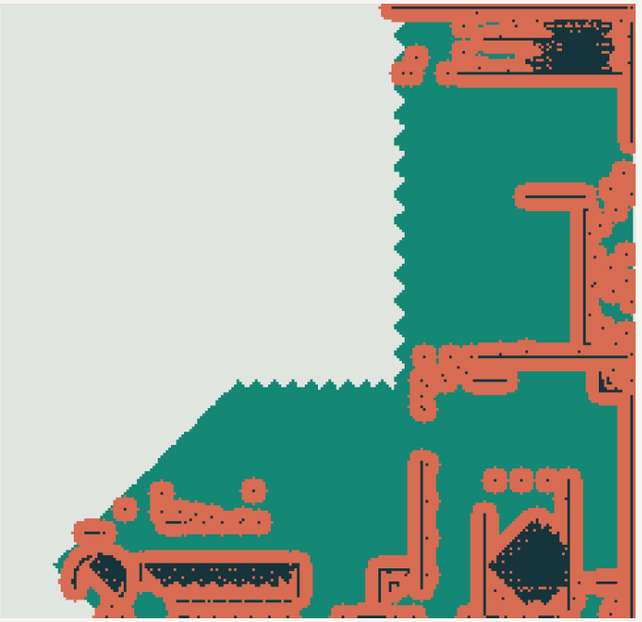
STEP 01 / 50

UNKNOWN 89.67%



STEP 25 / 50

UNKNOWN 69.27%



STEP 50 / 50

UNKNOWN 46.28%

UNKNOWN: 89.67% to 46.28%

FREE: 9.59% to 30.42%

The L-shaped path is simulated. State changes include obstacle inflation and accumulated evidence; non-UNKNOWN share is not a validated exploration-coverage metric.

S2-S3: 50 saved arrays and generated trajectory | Percentages recomputed from saved arrays; algorithm not rerun.

Make the map a contract.

An explicit representation makes a prototype inspectable and easier to pass between modules.

FOUR-STATE PROTOTYPE / SAVED ARRAY

234 x 242

uint8 cells / 0.05 m per cell

56,628 bytes

55.30 KiB of state-array payload.
One byte per cell.

Excludes file headers, point clouds, count grids and runtime memory. This is not a measured embedded RAM requirement.

STATE ENCODING

0	UNKNOWN
1	FREE
2	OCCUPIED
3	INFLATED

COORDINATE CONTRACT

```
ix = floor((x - x_min) / res)
iy = floor((y - y_min) / res)
```

```
state[iy, ix]
```

Metadata records the origin, resolution, axis convention, state meanings and cell-centre conversion.

Prototype inflation radius: 0.20 m + 0.05 m margin. This setting belongs to the early prototype, not the final robot footprint.

S2-S3: grid_meta.json, build_state_map() and saved uint8 arrays | Payload derived as 234 x 242 x 1 byte.

Define the next test as carefully as the result.

The current evidence supports a feasibility study. Deployment needs a different level of validation.

EVIDENCE IN HAND	ENGINEERING IMPLICATION	NEXT VALIDATION
Sparse observations Single-frame scenes retain large UNKNOWN regions.	Treat unknown space explicitly; do not infer clearance from absent points.	Compare against registered reference geometry; measure free-space false positives and retained unknowns.
Ray origin & visibility The Scanmap prototype casts rays from each sensor position to merged obstacle candidates.	Keep point-to-sensor provenance and first-hit visibility before trusting free-space rays.	Use occluding-wall cases with known poses to check that hidden cells remain unobserved.
Physical model & resources The early prototype uses 0.25 m inflation; no personal MCU timing or RAM benchmark is recorded.	Bind clearance to the actual platform and measure the complete pipeline.	Test narrow passages with the real footprint; profile memory and latency on target hardware.

S1: reported limitations | S2-S4: prototype source review | Next-validation items are proposed work, not completed results.

A result should have a traceable origin.

Source keys connect every claim to project documents, saved outputs or explicitly marked calculations.

S1 **Final report / personal study and team attribution**

FinalReport.pdf: contribution pp. 46-48; Appendix D pp. 52-59; Tables 4-7 pp. 58-59. PDF page numbers. Chi Zou is credited for the experimental mapping study.

S2 **Early prototype implementation**

Achievement/pythonfile/Projector/projector.py; Projector with car/projectorwithcar.py. Ground alignment, view simulation, four-state grids and metadata export.

S3 **Saved simulation evidence**

Achievement/pythonfile/Projector with car/: frames/trajectory_xy.txt; maps/grid_state_0000-0049.npy; grid_state_final.npy; grid_meta.json. Portfolio statistics are recomputed from these arrays.

S4 **Coordinate and ray-tracing prototype**

Achievement/pythonfile/Scanmap/maptransfer.py; maps/grid_meta_global.json; Achievement/Morefigure/coordinate.txt. Source inspection supports the visibility limitation on p. 9.

S5 **Individual process record**

Progress Report/progress_report_zc.docx; Project Proposal/proposal.docx; Final Report/Part4formal.docx. Used to distinguish technical exploration, experiment records and planned work.

S6 **Original figure**

Presention/5d semantic and cost maps.jpg. Saved semantic and cost-map figure on p. 3; no synthetic experimental imagery is used.

Team credit. Zhuyu Dong: robot platform/control and final mapping refactoring/integration. Junjie Ouyang: image-to-point-cloud reconstruction. Qingzhou Zhang: point-cloud merging/fusion. Weijie Chen: path planning.

Reproducibility status. The reported 13-scene study is documented, but its newmap.py and matching full inputs are absent from this local archive. Saved prototype arrays were rechecked; the full algorithms were not rerun. No external ground truth is available for the reported study.